

ATAK Development Environment Setup

UNCLASSIFIED

Prerequisites

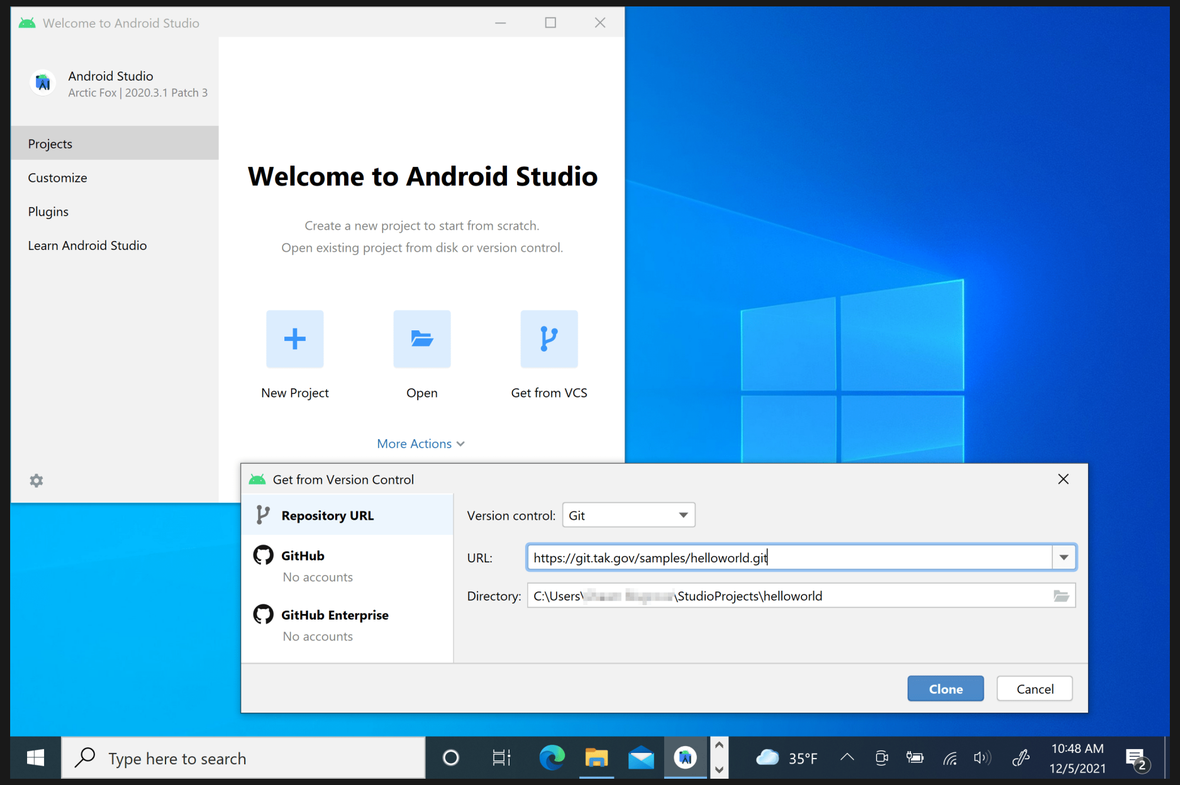
The following tools should be installed before proceeding with the setup in this document:

- Adoptium Java JDK 11 **DO NOT USE ORACLE**
- Git 2.34.1 or later
- Git Large File Storage (LFS) 3.0.2 or later
- Git Client (optional)
- Android Studio Dolphin or later

Testing an Existing Project

Let’s start with an existing plugin to demonstrate the mechanics of how to use Android Studio to checkout, build, and deploy a plugin. This information will be repeated later for the plugin that we will begin to modify. First familiarize yourself with the checkout process.

The URL you will use when cloning is here



Testing an Existing Project (Hello World)

Please note that if you have an issue with cloning you may need to take an extra step to enable the credential helper in Android Studio stack overflow help Effectively you need to enable the Credential Helper Settings > Version Control > Git In the bottom, enable Use credential helper if disabled.

When the project opens up, it will not be able to build initially. The next step is critical. We need to add information into the *local.properties* file at the root of the plugin project and add the following lines. Each of the <> entries needs to be replaced. Once completed, select **File → Sync Project with Gradle Files**.

```
sdk.dir=<directory-path-to-the-sdk>
takrepo.url=https://artifacts.tak.gov/artifactory/maven
takrepo.user=<username>
takrepo.password=<password>
takdev.plugin=.
```

Additional File [temporary]

- Download the appropriate Javadoc for the version being developed against. In this example we are pulling down the javadoc from here. The jar file needs to be placed in the root of the plugin directory. This is the case for each plugin you have.
- If you need a specific version other than the latest, you can download it from the atak-civ SDK repo.

Ways to run Android Applications

- There are two ways to run Android applications.
- Each one has advantages and disadvantages.
- You only need to set up one.
    - Physical device
    - Emulator

Configure a Physical Device

- Make sure phone has Android API 21 or later.
- Connect your device to the laptop, using a USB cable.
- Enable USB debugging
    - Open **Settings** app
    - Scroll to bottom and select **About phone**
    - Scroll to bottom and tap **Build number** seven times
    - Return to previous screen and click **Developer options**
    - Enable **USB debugging**

Configure an Emulator

- Select **Tools → AVD Manager** or on newer Android Studio releases **Tools → Device Manager** followed by **Create Device**

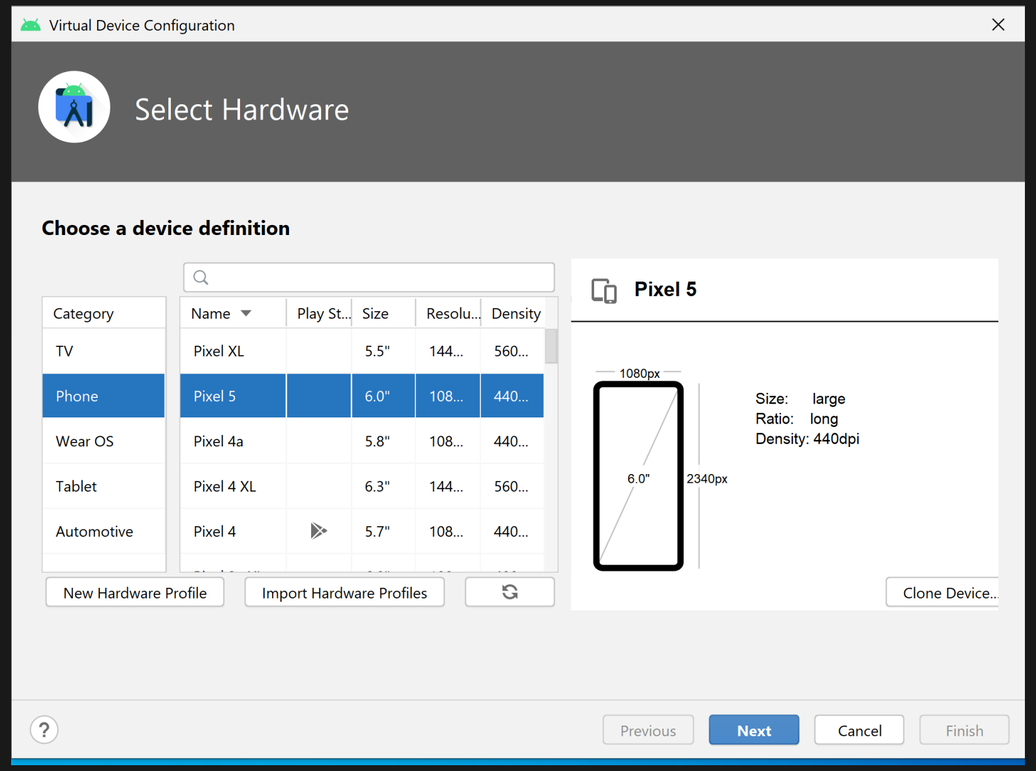
New Virtual Device

- Click **Create Virtual Device…**



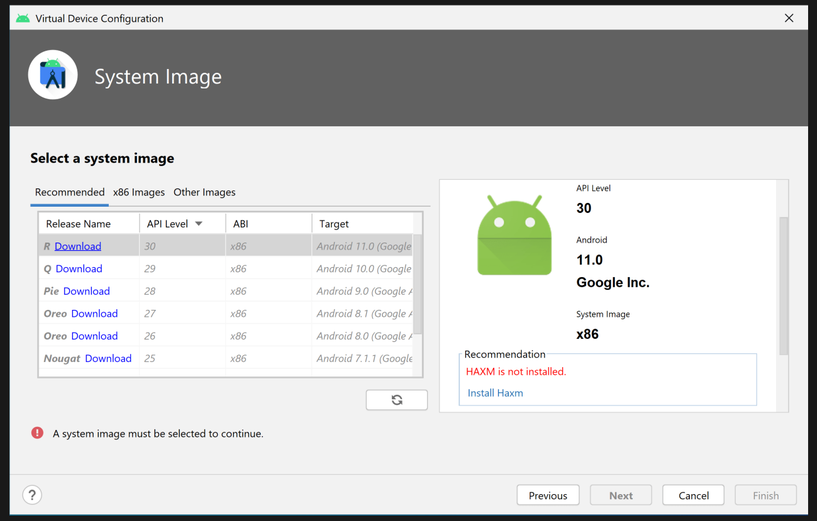
Configure an Emulator

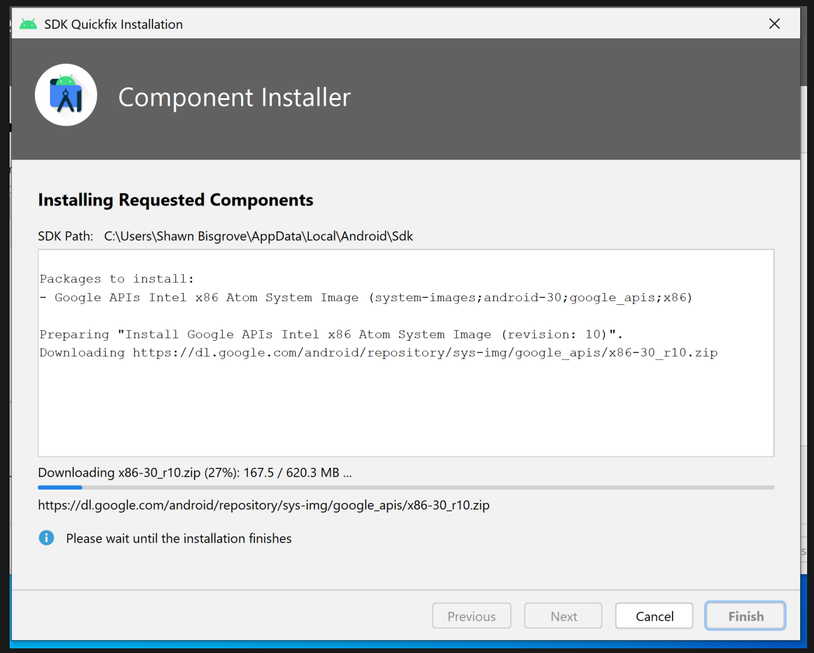
- Select **Pixel 5** then click **Next**



Configure an Emulator

- Select the **x86 Images** tab
- Click **Download** next to API 30 (R - ABI x86_64)
- Accept the license agreement and click **Next**
- After the download completes, click **Finish**





ATAK will run on anything above API 21 but it is recommended to run a newer version of Android.

Configure an Emulator

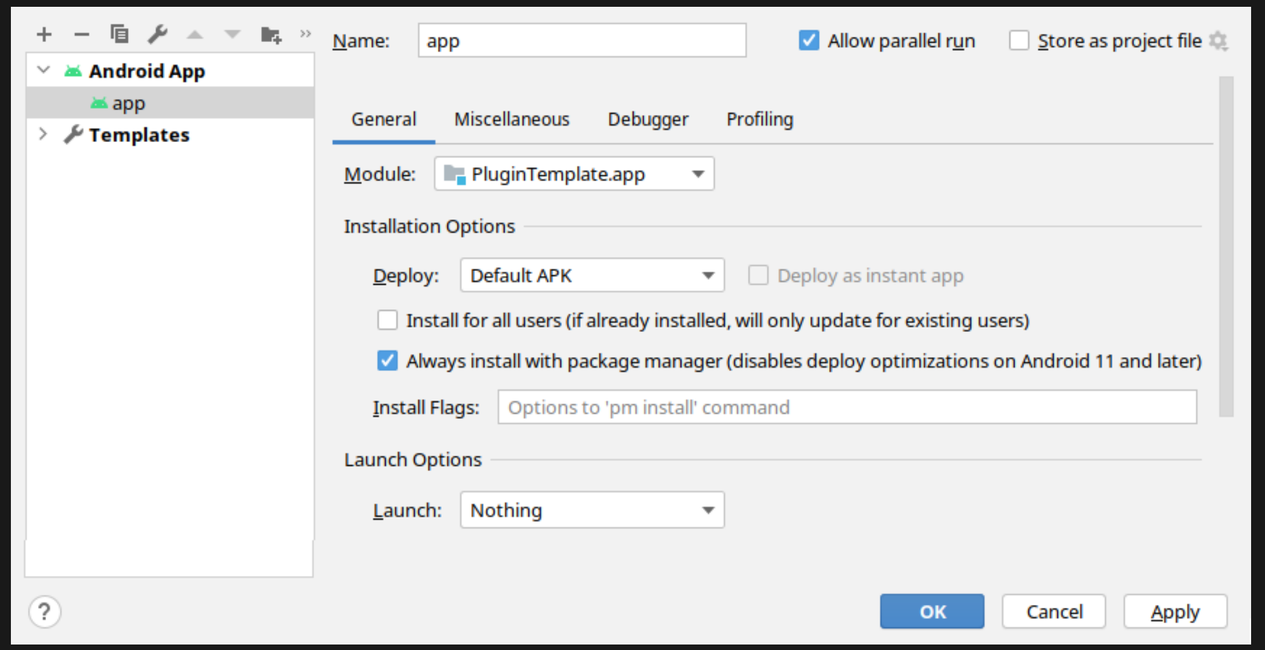
- Once downloaded, select the **API 30** image and click **Next**
- Name your emulator.
- Click on **Show Advanced Settings** and scroll down to adjust the VM Heap and Storage if they are less than the values below.
    - **VM heap**: 512 MB
    - **Internal Storage**: 2048 MB
- Click **Finish**
- You may get a warning about installing HAXM if running on Windows or Mac. At the time of these instructions, installing HAXM from Android Studio does not work. Install the latest release from the HAXM Github

Configure an Emulator

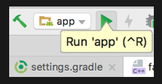
- To open the emulator, click the **play** button
- A phone should appear and boot up to the Android home screen

Verify the system is set up properly (VM and Physical Devices) 1/2

- In the main development window, go under **Build → Select Build Variant** and select **civDebug**
- Under the app pull down, select **Edit Configuration** and make sure **Launch Options** indicates Nothing as well as under **Installation Options** indicates Always Install with Package Manager. Both of these options are important



Verify the system is set up properly (VM and Physical Devices) 2/2

- Close the window and apply changes.
- Press the **play** button. 

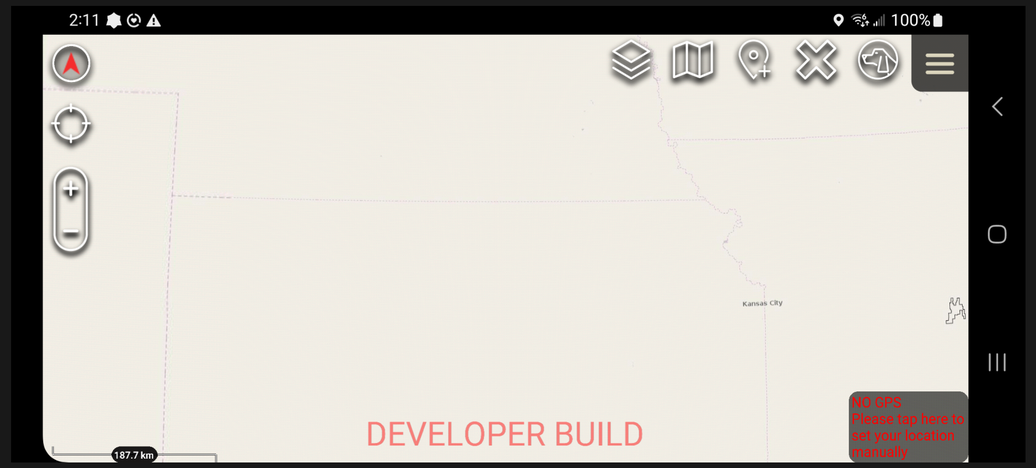
- Now go under the root of the plugin folder and go to */app/build/outputs/atak-apks/sdk/* using your favorite file browser. The single apk in this directory is the **ATAK** application that needs to be installed.
    - You can drag and drop this file over to the running VM to install it or if you are on a physical device you can use either ADB or drag and drop it onto the device and double click on it using the device file browser.

Install the Javadocs

- On the **Project** panel, select **Project** view from the drop-down at the top
- Under External Libraries, right-click on **Gradle:com.atakmap.app.civ.common:api:X.X.X.X** and select library property
- Select the **+** button and navigate to the plugin directory and select *atak-javadoc.jar*
- Note: If you do not see any JavaDocs when hovering over existing methods, you might need to turn on Quick Documentation:
    - Navigate to **File→Settings** and select **Editor → Code editing**
    - Check **Show quick documentation on mouse move**

Running ATAK

- Click on the ATAK icon in either the VM or on the physical device. (Note: If it crashes immediately - continue on to the next screen)
- Accept all permissions. View and Agree to the EULA.
- Once running hit **play** on your plugin development screen which will compile and load your plugin. ATAK will prompt you to load the plugin.
- Each time you hit the **play** button in your plugin IDE environment while ATAK is running, it will prompt you to load the new plugin. This will unload the previous version and load the new version. Resource management is critical. So anything you register should be unregistered during unloading; `oncreate`, `onstart`, `onpause`, `onresume`, `onstop`, and `ondestroy` should be considered.



ATAK (Running)



Install ATAK - Troubleshooting Tips

Problem	Potential Solution(s)
Out of space error.	Check your emulator advanced settings to see if your heap is set to 512 MB, and internal storage to 2048 MB.
ATAK keeps crashing.	This is possibly due to the fact that ATAK makes use of a significant amount of OpenGL capability that your system might not support in emulation. ▪ Click on the three dots at the bottom of the emulator options. ▪ Navigate to Advanced… ▪ Try out a few shader options from the dropdowns, and restart the emulator. ▪ If none of those work you may need to upgrade graphics drivers or run with a real hardware device.

Install ATAK - Troubleshooting Tips

⚠ If the apk fails to install, execute the following command in the current directory to install the apk, substituting the actual name of the ATAK apk for <atakApk>:

Windows

```
C:\Users\%USERNAME%\AppData\Local\Android\Sdk\platform-tools\adb.exe install <atakApk>
```

Mac

```
/Users/${USER}/Library/Android/sdk/platform-tools/adb install <atakApk>
```

Install ATAK - Troubleshooting Tips

⚠ If ATAK crashes repeatedly, execute the following commands, which will create a file on the emulator to instruct ATAK not to use OpenGL

Windows

```
cd C:\Users\%USERNAME%\AppData\Local\Android\Sdk\platform-tools
adb.exe shell
cd sdcard/atak/
touch opengl.broken
exit
```

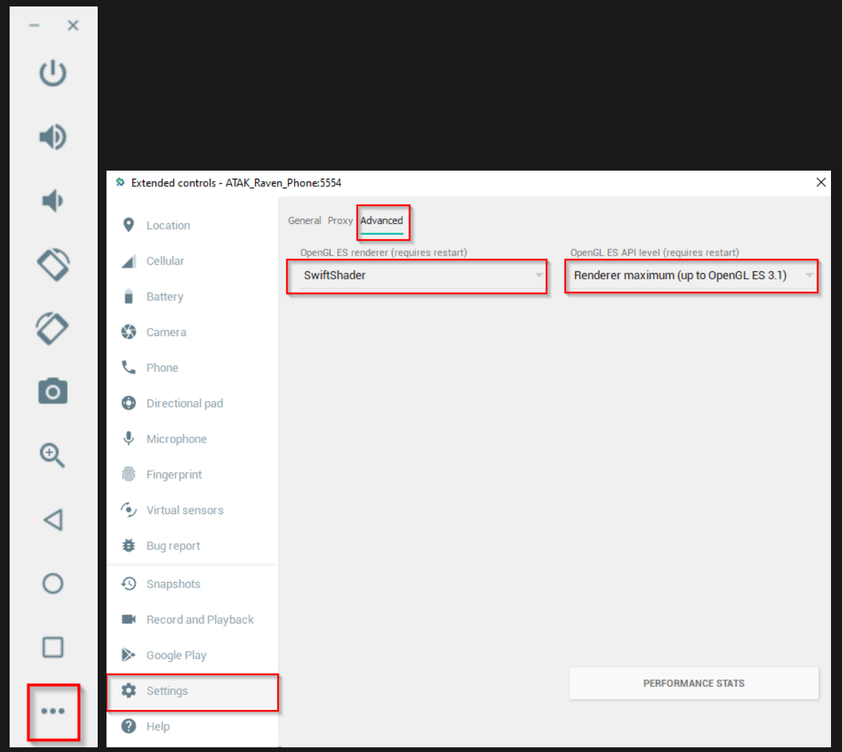
Mac

```
cd /Users/${USER}/Library/Android/sdk/platform-tools
adb shell
cd sdcard/atak/
touch opengl.broken
exit
```

Install ATAK - Troubleshooting Tips

⚠ If ATAK still crashes, has garbled display text, or just displays a black screen, follow these instructions to adjust the renderer settings.

- Click the icon on the emulator controls
- Select Settings in the dialog
- Select the Advanced tab
- Change OpenGL ES renderer to ANGLE (D3D11) or SwiftShader
- Change OpenGL ES API level to Renderer maximum
- Restart your emulator (using the X button in the emulator controls)



Create a New Project

We always start a new project from the plugin template.

- Pull down the PluginTemplate as a zip from here
- Unzip the file and rename the PluginTemplate directory to the name of your plugin.
- In the root of the directory, rename the file **template.local.properties** to **local.properties** and modify the file appropriately

Create a New Project

- Open Android Studio
- Select open project
- Navigate to location that the Plugin Template was unzipped
- Open the root of the project
- Setup the *local.properties* just as before.
- Begin development!